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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/591,025	02/02/2022	Rémy Orban	2019P00556 US (G146US2)	2893

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EXAMINER	
HAWKINS, DOMINIC E	

ART UNIT	PAPER NUMBER
2858	

NOTIFICATION DATE	DELIVERY MODE
01/18/2023	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 17/591,025	Applicant(s) Orban et al.	
	Examiner DOMINIC E HAWKINS	Art Unit 2858	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on February 02, 2022.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-18 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1 is/are rejected.

8) ☑ Claim(s) 2-18 is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 02/02/2022 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☑ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 02/02/20222.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Detailed Action

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

2. Claims 1-18 of U.S. Application 17/591,025 filed on February 02, 2022 are presented for examination.

Information Disclosure Statement

3. The information disclosure statement (IDS) submitted on 02/02/2022 has been considered by the examiner.

Priority

4. Receipt is acknowledged of certified copies of papers submitted under 35 U.S.C. 119(a)-(d), which papers have been placed of record in the file.

Claim Rejections - 35 USC § 101

35 U.S.C. 101 reads as follows:

Whoever invents or discovers any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof, may obtain a patent therefor, subject to the conditions and requirements of this title.

3. Claim 14 is rejected under 35 U.S.C. 101, as being indefinite for failing to particularly point out and distinctly claim the subject matter which applicant regards as the invention.

The statutory categories under 35 U.S.C. 101 are expressed in the alternative, i.e., process, machine, manufacture or composition of matter. Thus, claim 14, which encompass two statutory categories, i.e., apparatus (an electrical switching apparatus) and method for operation, violate 35 U.S.C. 101. *See Ex parte Lyell*, 17 USPQ2d 1548, 1551 (Bd. Pat. App. & Int. 1990).

Moreover, there are two separate requirements set forth in the second paragraph of 35 U.S.C. 112:

(A) the claims must set forth the subject matter that applicants regard as their invention; and

(B) the claims must particularly point out and distinctly define the metes and bounds of the subject matter that will be protected by the patent grant.

It is unclear whether claim 14 is claiming an apparatus or a method. Claim 14, therefore, are indefinite in violation of 35 U.S.C. 112, second paragraph. *See Lyell*, 17 USPQ2d at 1552.

Firstly, with regard to claim 14, this claim do not set forth any steps involved in the method/ process, it is unclear what method/ process applicant is intending to encompass (A method for estimating the state of an electrical switching apparatus as claimed in claim 1).

A claim is indefinite where it merely recites a use without any active, positive steps delimiting how this use is actually practiced. Without setting forth any steps involved in the process results in an improper definition of a process, i.e., results in a claim which is not a proper process claim under 35 U.S.C. 101. Applicants are advised to follow a proper method claim format that

conforms to current U.S. practice (For example, "A method for/of .... comprising steps of:" and ***a series of active, positive steps*** involved in the process) (emphasis added).

The essential purpose of patent examination is to determine whether or not the claims are precise, clear, correct, and unambiguous to ensure that the scope of the claims is clear so the public is informed of the boundaries of what constitutes infringement of the patent. Therefore, the uncertainties of claim scope should be removed as much as possible.

Claim Rejections - 35 USC § 103

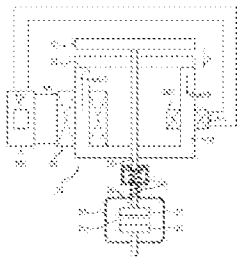
5. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102 of this title, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the

statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

6. Claim 1 is rejected under 35 U.S.C. 103 as being unpatentable over Hahn et al (USPGPub 20160293364) in view of Hajati et al (USPGPub 20080316296).



Prior Art: Hahn

7. Regarding claim 1, Hahn discloses a method for estimating the state of an electrical switching apparatus (20), the switching apparatus comprising separable electrical contacts set up to be coupled to an electrical conductor (not fully shown but I(t) comes from 100 therefore there has to be a conductor and contacts to the coil. Also since this is a method claim and these components does not necessarily involve in the steps/method of performing testing), and an electromagnetic actuator (10), controlled by a control circuit (110) and comprising a movable core (60 including 50) , coupled to the separable contacts, and a coil (80 and 200) , which is passed through by a coil current (It), wherein the method comprises: measuring, by means of sensors (using Hall 400) associated with the coil, values of the coil current during a time interval (time profile), while the electromagnetic actuator is kept in a stable position (such as position 61), in particular a closed position (par 33 discloses being in closed position); computing, by

means of an electronic control device (100) of the switching apparatus, one or more values of a magnetic flux of the coil from the measured coil current values (claim 1 and abstract discloses at least voltage). Hahn does not fully disclose using previously known coil voltage and coil resistance values.

However, Hajati discloses using previously known coil voltage and coil resistance values (abstract and claim 13 disclsoes measuring different characteristics of the actuator based on voltage and coil resistance). It would have been obvious to a person having ordinary skill in the art at the time the invention was made to have provided combined Hahn in view of Hajati in order to determine the voltage of the switch to engage the actuator.

Allowable Subject Matter

8. Claims 2-18 are objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims.

9. Regarding claim 2, the prior art of record taken alone or in combination fail to teach or suggest a method for estimating the state of an electrical switching apparatus, the switching apparatus comprising separable electrical contacts set up to be coupled to an electrical conductor, and an electromagnetic actuator, controlled by a control circuit and comprising a movable core, coupled to the separable contacts, and a coil, which is passed through by a coil current wherein the method comprises wherein the one or more values of a magnetic flux ~~(0)~~ are computed by integrating the following formula:

$$\phi = \int \frac{U_{BOB} - R_{BOB} \cdot I_{BOB}}{N_{BOB}} dt \qquad \text{~~[Math 5]~~}$$

where 0 is the magnetic flux,

UBOB is the previously known coil voltage, RBOB is the coil resistance, IBOB is the coil current, and NBOB is a number of turns of the coil in combination with the other limitations of the claim.

10. Regarding claim 3, the prior art of record taken alone or in combination fail to teach or suggest a method for estimating the state of an electrical switching apparatus, the switching apparatus comprising separable electrical contacts set up to be coupled to an electrical conductor, and an electromagnetic actuator, controlled by a control circuit and comprising a movable core, coupled to the separable contacts, and a coil, which is passed through by a coil current wherein the method comprises wherein a first value of the magnetic flux is computed from measurements of the coil current taken while the switching apparatus is in a freewheeling mode, in which the coil voltage is passively limited by the control circuit, while the coil current decreases freely, with the electrical energy being mainly dissipated by the coil resistance in combination with the other limitations of the claim.

11. Claims 4-7 are also objected to as they depend on objected claim 3.

12. Regarding claim 8, the prior art of record taken alone or in combination fail to teach or suggest a method for estimating the state of an electrical switching apparatus, the switching apparatus comprising separable electrical contacts set up to be coupled to an electrical conductor, and an electromagnetic actuator, controlled by a control circuit and comprising a movable core, coupled to the separable contacts, and a coil, which is passed through by a coil current wherein the method comprises wherein the one or more computed values of magnetic flux are normalized

and combined with at least one other measured and/or computed electrical quantity of the switching apparatus, with each other electrical quantity measurement also being normalized, so as to arrive at an intermediate indicator of the state of the electromagnetic actuator in combination with the other limitations of the claim.

13. Claims 9-13, and 15-18 are also objected to as they depend on objected claim 3.

14. Regarding claim 14, the prior art of record taken alone or in combination fail to teach or suggest a method for estimating the state of an electrical switching apparatus, the switching apparatus comprising separable electrical contacts set up to be coupled to an electrical conductor, and an electromagnetic actuator, controlled by a control circuit and comprising a movable core, coupled to the separable contacts, and a coil, which is passed through by a coil current wherein the method comprises with the switching apparatus being set up to be coupled to an electrical conductor and comprising: separable contacts moved between an open position and a closed position by an electromagnetic actuator comprising a coil and a movable core linked to the separable contacts in combination with the other limitations of the claim.

Prior Art

The prior art made of record and not relied upon is considered pertinent to applicant's disclosure.

Kim et al (US Pat No. 9293243): discloses electromagnetic switching device.

Lammers et al (US Pat No. 6262648): discloses actuator for moving a contact into on and off state.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to DOMINIC E HAWKINS whose telephone number is (571)272-2647. The examiner can normally be reached Monday-Friday 7:30am-5:00pm EST. Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Judy Nguyen can be reached on (571) 272-2258. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/DOMINIC E HAWKINS/

Primary Examiner, Art Unit 2858